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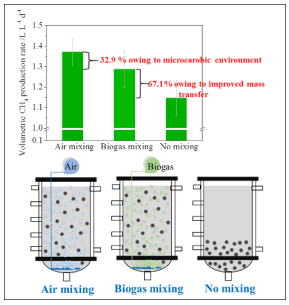
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Using air instead of biogas for mixing and its effect on anaerobic digestion of animal wastewater with high suspended solids

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GRAPHICAL ABSTRACT



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ABSTRACT

This study proposed a new mixing method for anaerobic digestion treating animal wastewater using air as gas source of agitation (named "air mixing") and demonstrated its feasibility by comparing with other mixing modes. The results indicated that the methane production for air mixing was increased by 6.4%, 11.9% and 19.6% compared with biogas mixing, mechanical mixing and no mixing. Air mixing improved the mass transfer and the homogeneous mixing time was shortened from 10 min of mechanical mixing to 1.5 min at the same power input. A transient microaerobic environment was created by air mixing, which increased the hydrolysis efficiency by 1.7–11.4% compared with biogas mixing and facilitated VFAs generation and consumption, as well as promoted the syntrophic relationship between facultative bacteria and hydrogenotrophic methanogens. The relative contribution of the improvement of mass transfer and the reaction of microaerobic environment to methane production was 62.9% and 37.1%, respectively.

1. Introduction

Anaerobic digestion (AD) is an effective method for the treatment of animal wastewater (Yang et al., 2019). However, the presence of high suspended solids (> 2000 mg L^{-1}) in the wastewater causes the difficulty of mass transfer in the system and the formation of scum and sediment, and negatively impacts functioning and performance of anaerobic reactors (Zeng et al., 2019). Proper mixing during anaerobic treatment is considered to be an effective method to increase biomass conversion efficiency and methane yield because mixing distributes substrates, enzymes, microorganisms and temperature throughout the digester. The importance of mixing has been reported by many researchers (Kariyama et al., 2018; Singh et al., 2019). From a detailed review of the digestion process and commercial large-scale digesters, the mixing of an anaerobic digester is accomplished through a motor-driven mixer (i.e., mechanical mixing), recirculation of the mixed

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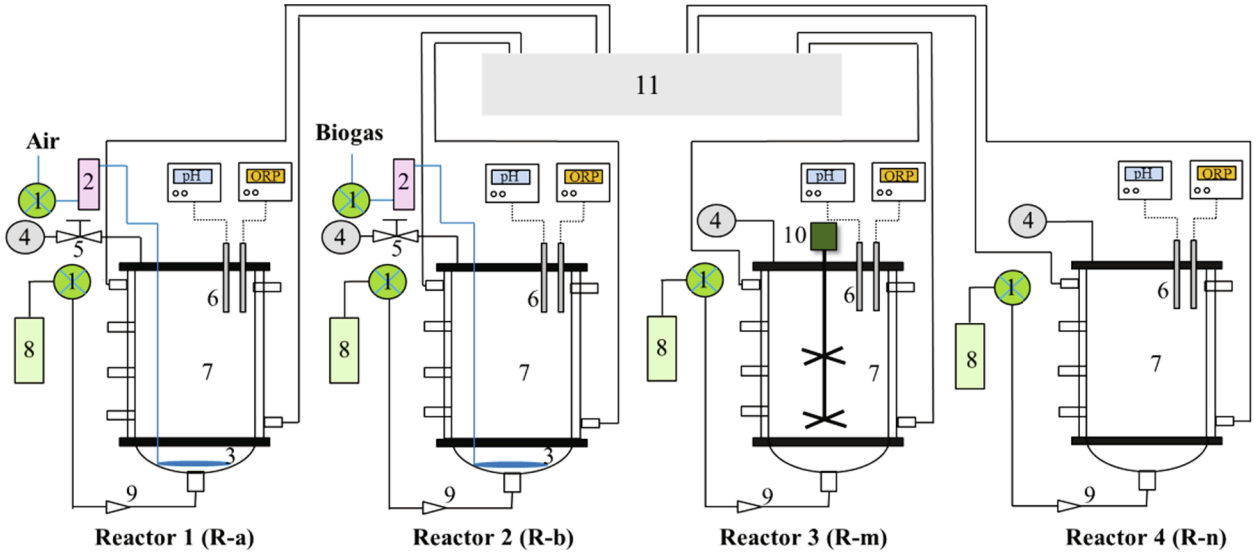


Fig. 1. Schematic diagram showing components of the reactors.

liquor (i.e., hydraulic mixing) or biogas (i.e., biogas mixing) (Kariyama et al., 2018).

Advantages and disadvantages of various mixing methods in anaerobic digesters are well documented in the literature (Latha et al., 2019; Singh et al., 2019). Mechanical mixing has been proved to be more economical than recirculation in terms of power consumption per volume mixed (Singh et al., 2019); however, the mechanical parts inside the digester are not easily accessible for maintenance (Latha et al., 2019). Hydraulic mixing is more flexible and less prone to gas leakage than mechanical mixing, but has difficulty mixing the substrate and sludge thoroughly. "Short circuits" and "dead spaces" where hydraulic mixing is inadequate are common in digesters (Bhattacharjee et al., 2015). Therefore, hydraulic mixing is generally only suitable for small reactors that treat wastewaters having low solids concentrations (Karim et al., 2005). Biogas mixing achieves good mass transfer and heat transfer at low energy consumption and uses a simple configuration that has no moving parts inside the digester. It is generally recognized that biogas mixing is superior to other modes of mixing in terms of energy requirement, operation costs and reliability over long-term operation (Kuczman et al., 2017; Latha et al., 2019). However, biogas mixing is not frequently used in on-farm biogas plants because it creates difficulty in maintaining sufficient mixing when the biogas production is low. In addition, relatively expensive explosion-proof motors and other safety devices are needed when pumping biogas for mixing.

It is reasonable to question whether air instead of biogas is used for mixing the contents of anaerobic digesters, thereby resolving the problems of insufficient biogas production and safety. So far, no research has been reported about using ambient air as the gas source for mixing in AD. In fact, exposing AD systems to oxygen (or air) traditionally has been avoided because of its negative effects on the growth and activity of obligate anaerobes, especially methanogens, by damaging the cell membrane, crucial proteins, and other cell constituents (Bhattad et al., 2017). However, it has been found in recent years that strict anaerobes have several mechanisms that allow them to tolerate oxygen or air and experience few or no inhibitory effects (Botheju and Bakke, 2011; Nguyen and Khanal, 2018). Numerous researchers have applied microaeration in the AD process, which is achieved by injecting a limited amount of air or oxygen into the digester to enhance hydrolysis, control accumulation of volatile fatty acids (VFAs) and remove hydrogen sulfide (H_2S) (Krayzelova et al., 2015; Nguyen and Khanal, 2018; Nguyen et al., 2019; Ruan et al., 2019). However, these studies injected air/oxygen into the headspace of digesters or into the liquid mixture at a low gas flow rate in the range 3–10 mL/L/min (Nguyen and Khanal, 2018). In other words, the purposes of air/oxygen injection into the digesters in previous studies were not for mixing, but rather for producing a microaerobic environment.

In this research, the novel idea of using air instead of biogas as a gas source for gas mixing (named "air mixing") was studied with the objective of establishing a new mixing technology. To achieve this objective, several scientific questions had to be answered. First, can air mixing achieve adequate agitation without causing a substantial negative impact on the anaerobic environment for methane production? Second, what is the longest continuous air injection time that can be employed without affecting the anaerobic environment for methane production? Third, if air mixing is feasible and increases the efficiency of anaerobic digestion, what are the relative contributions of mixing and a microaerobic environment?

Therefore, the overall goal of this study was to assess the feasibility of air mixing and evaluate its effects on the AD process based on the methane production efficiency. By combining reactor performance and microbial community analyses, new insights were sought into the response of AD to air mixing so that an effective new mixing strategy for increasing the methane production of an AD system could be developed.

2. Material and methods

2.1. Reactor

Four, 4.5 L anaerobic sequencing batch reactors, each with a working volume of 4.0 L (11 cm internal diameter, 45 cm liquid height), were used (Fig. 1). Reactors 1 (R-a) and Reactors 2 (R-b) were mixed by air and biogas, respectively. The gas was introduced into the reactors via a digital peristaltic pump and an aeration disc at the bottom of the reactors. Reactor 3 (R-m) was mixed using a 6-cm diameter axial-flow impeller (NZ-1300, Eyela, Tokyo, Japan). Reactor 4 (R-n) was operated as blank control without any mixing. Each reactor was equipped with three sampling ports at different heights along the side of the reactor, as shown in Fig. 1.

Table 1
Overview of steady-state process parameters, volumetric CH_4 production rates, biogas composition and volatile fatty acids (VFAs) concentrations in anaerobic reactors with different mixing methods.

Reactor	Volumetric CH_4 production rate (L/L/d)	Methane yield (L_{CH4}/gVS)	Biogas composition					VFAs concentrations (mg/L)		
			CH_4 (%)	O_2 (%)	H_2S (ppm)	CH_4/CO_2 ratio	Acetate	Propionate	TVFA	A/P ratio
R-a	1.371 ± 0.067	0.229 ± 0.010	53.8 ± 1.5	2.3 ± 0.1	100 ± 34	1.59 ± 0.07	296 ± 177	44 ± 11	351 ± 186	6.27 ± 2.31
R-b	1.288 ± 0.059	0.215 ± 0.010	60.6 ± 1.9	1.1 ± 0.2	271 ± 82	1.77 ± 0.15	262 ± 167	46 ± 19	313 ± 179	5.66 ± 2.56
R-m	1.225 ± 0.059	0.205 ± 0.010	60.9 ± 1.5	0.9 ± 0.4	293 ± 103	1.74 ± 0.14	197 ± 82	46 ± 22	250 ± 101	4.45 ± 0.44
R-n	1.147 ± 0.058	0.191 ± 0.010	60.2 ± 2.1	0.9 ± 0.5	336 ± 150	1.70 ± 0.07	366 ± 352	73 ± 48	434 ± 343	4.42 ± 1.36

2.2. Substrate and inoculum

Swine wastewater was used as the substrate and was collected from a farm located in Jianyang, Sichuan, China. Samples were transported back to the laboratory immediately after collection and stored at 4 °C prior to use. The swine wastewater was diluted with a certain amount of tap water to achieve a total solids (TS) concentration of 6%. The inoculation sludge for the AD experiments was harvested from a laboratory-scale mesophilic anaerobic digester that treated swine wastewater (Supporting Information).

2.3. Comparison experiments of air mixing with other mixing modes

To begin the experiments, 2 L of inoculum sludge was added to each reactor. The four reactors were fed swine wastewater once per day (semi-continuous mode) at a loading rate of 6.0 g VS/L/d. The steady state was defined to be the point when biogas production varied within 10% of the mean between feeding times within 2–3 weeks. Each reactor was maintained at 35 ± 1.0 °C by an electrically heated water jacket surrounding the reactor. Biogas was collected in aluminum gas-bags for subsequent analysis.

The three reactors with mixing (R-a, R-b and R-m) were agitated three times per day for 1.5 min on each occasion: once immediately after feeding, once 2 h after feeding, and once 6 h after feeding. All three reactors were mixed at the same power input per unit volume of mixed liquid condition (i.e., 5 W/m^3). The gas flow rate for R-a and R-b and the impeller speed for R-m were calculated as described by Karim et al. (2005). Detailed calculation is given in supporting information. A gas flow rate of 66.7 mL/L/min was supplied to reactors R-a and R-b and the R-m was operated at 88 rpm.

2.4. Degree of mixing

The degree of mixing of the digester contents was reflected by the TS distribution within the reactors, which was determined by collecting samples for TS analysis from three different heights (10, 20 and 30 cm) above the base of the digesters after mixing. The coefficient of variance (κ) was defined by the ratio of standard variation of TS concentrations to average of TS concentrations as described in Eq. (1) and was used as a measure of degree of mixing. Furthermore, "homogeneous mixing time" was defined as the time required achieving a certain degree of mixing. In this paper, a κ-value less than 0.02, was considered to indicate homogeneous mixing.

$$\kappa = \frac{\sqrt{\sum_{1}^{i} \frac{(C - C_{av})^2}{i}}}{C_{av}} \quad (1)$$

In Eq. (1), i is the number of samples, C is the TS concentration (g/L) and C_{av} is the average TS concentration (g/L).

2.5. Analytical methods

The pH, ORP, and methane yield were monitored daily; individual VFA and biogas composition were analysed weekly; the crude protein, crude fiber and crude fat contents were determined monthly. Details of the analytical methods are provided in the Supporting Information.

Comparisons among values obtained in four reactors were made by one-way repeated analysis of variance, followed by Dunnett's t test. A value of $p < 0.05$ was considered statistically significant.

2.6. Microbiological analysis

To assess the microbial community and population shift responding to mixing, sludge samples were collected from each reactor at the conclusion of the experiment. Total genomic DNA was extracted using a PowerSoil DNA Isolation Kit (MO BIO, Carlsbad, CA, USA) according to the manufacturer's protocol. The primer pairs 344F-915R and 338F-806R were used to amplify the archaeal 16S rRNA genes and bacterial 16S rRNA genes as described in Yang et al. (2019). Detailed method description can be found in the Supporting Information.

3. Results and discussion

3.1. Reactor performance in different mixing modes

3.1.1. Volumetric methane production rate

The volumetric methane production rate is defined as the methane production per volume of digester per day and is a good indicator of the digester efficiency. Mean values for the volumetric methane production rate in the four reactors were determined based on steady-state conditions and are summarized in Table 1. The volumetric methane production rates of R-a, R-b, R-m and R-n were 1.371, 1.288, 1.225 and 1.146 L/L/d, respectively. The total methane production was significantly greater for three reactors with mixing than the reactor without mixing, which implied that a short mixing time of only 1.5 min enhanced the AD process. Kaparaju et al.(2008) had a similar finding; they observed that compared with biogas production from continuous mechanical mixing, production from minimal mixing (10 min mixing before feeding) increased by 12.5%. Among the four reactors, the differences in total methane production were statistically significant (one-way ANOVA, $p < 0.05$). The total methane production for R-a was the highest and increased by 6.4%, 11.9% and 19.6% compared with R-b, R-m and R-n, respectively. It was evident that air mixing not only did not limit methanogenesis, but also promoted the production of methane.

The biogas production rates (mL/min) during the 24 h after feeding are shown in Fig. 2. In the three reactors with mixing systems, the biogas production rate remained high in the first 12 h, in contrast to that in the reactor without mixing. Furthermore, a significant increase in the rate of biogas production per min was observed after each of the three mixing periods because mixing promoted the mass transfer and the release of CH_4. The digesters were not mixed in the subsequent 12 h, and there was no difference in the biogas production rates among digesters R-b, R-m and R-n, but the biogas production rate of R-a was obviously greater than other three reactors. ORP profiles of the R-a (Supporting Information) represented that about 0.5 h of a micro-aerobic condition (ORP value in the range of 0 to −300 mV) was

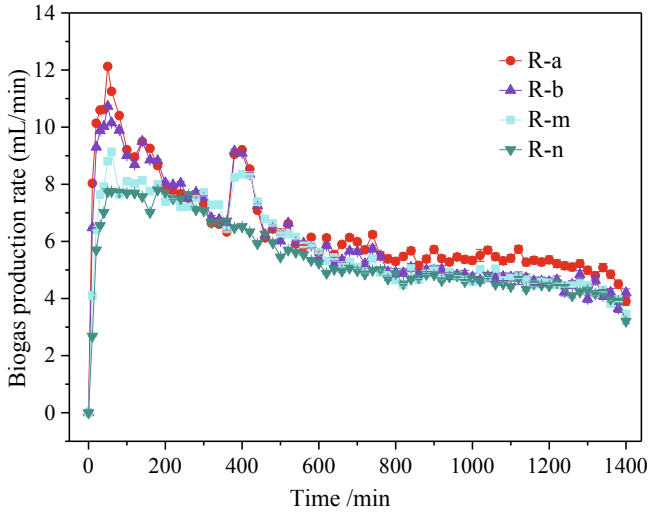


Fig. 2. Biogas production rate (per minute) during 24 h after feeding of the four anaerobic digesters.

creased after air mixing on each occasion. Therefore, the increased biogas production rate in R-a may have occurred because some refractory organic matter was converted to biodegradable ones in the transient microaerobic environment (Cheng et al., 2018; Chen et al., 2020).

3.1.2. Methane yield

Methane yield is the volume of methane produced per unit mass of VS added, and is an effective parameter by which to judge methane production efficiency in terms of loaded substrate. Previously, researchers reported that the injection of air into anaerobic digesters could result in some organic matter being consumed by the aerobic oxidation, thereby decreasing the methane yield (Díaz et al., 2010). Methane yields during the stable stage of AD are shown in Table 1 and were observed to be 0.229, 0.215, 0.205 and 0.191 L_{CH4}/gVS_{added}, for R-a, R-b, R-m and R-n, respectively, at VS solid loading of 6 g L^{-1} d^{-1}. The methane yield of R-n is similar to the results from AD experiments using swine manure as the substrate gave specific methane productions not exceeding 0.2 L_{CH4}/gVS_{added} (Borowski et al., 2014; Hu et al., 2019) when the loading rate exceeded 4 g VS L^{-1} d^{-1}. In this study, the mixing increased methane yield by 7.3%–19.9%. Compared to the methane yield of R-b and R-m, the methane yield of R-a increased by as much as 6.5% and 11.7%, respectively. These increases indicated that instead of decreasing methane yield, the transient microaerobic environment created by air-mixing in R-a actually improved the conversion of organic matter to CH_4 compared to that from the digesters that had other types of mixing.

3.1.3. Biogas composition

The mean values for biogas composition were determined based on steady-state conditions and are summarized in Table 1. The average CH_4 content in the biogas produced in R-b and R-m were not significantly different from that in biogas from R-n and ranged from 60.2% to 60.9%. By contrast, the average CH_4 content in biogas from R-a (53.8%) were lower by 10.6–11.6%. The CH_4 content in the outlet of the R-a declined to 33% after air mixing for 1.5 min as a result of the dilution by air used for mixing as shown in Fig. 3, and subsequently gradually increased and returned to its initial value of over 54% in 1.5 h. The injection air leaving the digester with the biogas into the gasbags represented a content of 10–12% in biogas composition as shown in Table 1. The CH_4/CO_2 ratio of biogas from R-a was 1.59, which was slightly lower than that in biogas from the other reactors (1.70–1.77) (Table 1). These results suggests that the aerobic oxidation of the

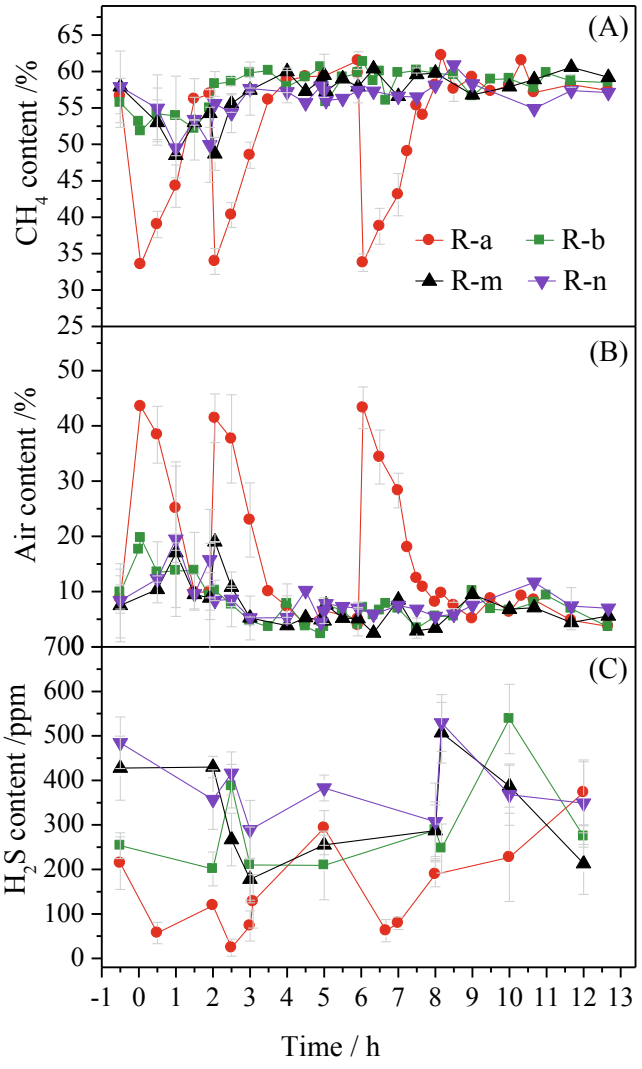


Fig. 3. Variations in biogas composition from four anaerobic digesters: (A) CH_4 content; (B) air content; (C) H_2S content.

organic matter to produce CO_2 was also a reason of the decrease in the CH_4 content observed in R-a.

The H_2S contents in R-a, R-b, R-m and R-n were 100, 271, 293 and 336 ppm, respectively (Table 1), and thus, the efficiency of desulfurization achieved by air mixing was 70.2%. This was significantly higher than the desulfurization efficiencies achieved by biogas mixing (19.3%) and mechanical mixing (12.8%), but not as high as reported (98–99%) in a previous study (Nghiem et al., 2014; Krayzelova et al., 2015). The almost complete desulfurization reported probably occurred because the technique used was like that of most microaerobic reactor methods, i.e., injecting air or O_2 directly into the headspace of the reactor where H_2S oxidation mainly takes place (Díaz et al., 2011; Krayzelova et al., 2015). By contrast, in the present study, air was injected directly into the liquid phase where the O_2 could have been consumed by other oxidative processes in addition to the oxidation of H_2S. In fact, the H_2S removal efficiency obtained in this study was similar to that reported by Krayzelova et al. (2014) of 73% who injected air into the liquid phase of an anaerobic digester. Although the efficiency of H_2S removal with air mixing was lower than the efficiency when air was applied in the headspace, but the decrease in the sulfide concentration in the liquid had an additional positive effect of decreasing the effect of sulfide toxicity on methanogens (Krayzelova et al., 2015). Fig. 3(C) shows the evolution of H_2S in biogas before and after mixing. The H_2S levels decreased after air injection (R-a) and then gradually increased, which

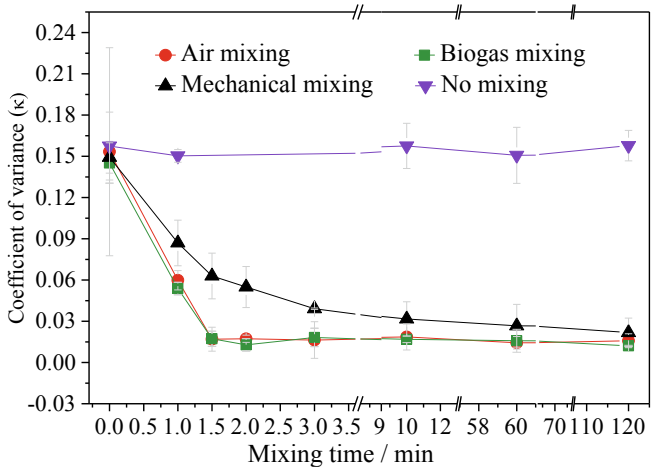


Fig. 4. Changes in the coefficient of variation (κ) of total solids (TS) concentrations in the reactor at different mixing techniques over mixing time.

indicated H_2S oxidation had stopped. Klok et al. (2012) reported that when ORP is lower than −420 mV, no sulfur can be obtained because the ORP is dominated by a high sulfide concentration and the biological reactions become inhibited. Therefore, shorter air injection time than other studies is also a main reason of low H_2S removal efficiency.

3.2. Mechanistic analysis of the positive effects of air mixing on AD

3.2.1. Improved mass transfer by mixing

As described in Section 2.3, the coefficient of variation (κ) of the TS concentrations at different depths in the digesters was used as a quantitative measure of the degree of mixing. A κ-value less than 0.02 (as calculated using Eq. (1)) indicated homogeneous mixing. Only the axial variation of TS concentrations was used to compare mixing performance because the high height-to-diameter ratio (4.2) of the digesters meant that axial mixing was a more time-consuming process than latitudinal mixing. Fig. 4 shows the variation of κ-values for different mixing techniques as a function of mixing time. At time (t) = 0 min, namely no mixing, TS was naturally distributed by gravity and many solids settled in the lower layer of the reactors. The κ-values for the different reactors were at their maxima (0.145–0.165) at this time. As mixing time increased, the substrate and sludge in the reactors were mixed and eventually TS became evenly distributed throughout the whole reactor. The κ-value in R-a decreased greatly with the increase of mixing time from 0 to 1.5 min and then approached a stable value of 0.016. The change of κ-values in R-b was not much different from that in R-a, which indicated that the homogeneous mixing time of both gas-mixed reactors were less than 1.5 min. However, the amplitude of the variation of κ-values in R-m was smaller than that for the two gas-mixed reactors and the time needed to reach a stable κ-value was about 10 min, which indicated that mechanical mixing required much longer to achieve homogenization than gas mixing. Mao et al. (2019) also reported a long time of 19 min required for an impeller-mixed digester to reach homogeneous mixing under similar a mixing condition.

The stable value of κ was about 0.025 in R-m and higher than the stable values in R-a and R-b (0.016 and 0.017, respectively), which may be due to the flow fields generated by the different mixing modes. In both gas-mixed reactors, bubbles released from the bottom of reactors mixed the digester contents by physically moving the substrate and sludge as the bubbles rose to the surface. The small gas bubbles also lowered the density and viscosity of the bulk solution; consequently, the digester contents in the gas-mixed reactors were easier to move and mix than those in the mechanically mixed reactor (Low, 2014). Previous research on radial-flow impellers for mechanical mixing showed that radial mixing was not totally effective for solid suspensions (Leonzio, 2019); although radial mixing was good, axial mixing was less so. Therefore, the results from the current study indicated that compared with no mixing, all three mixing modes improved the mass transfer effect. Moreover, both gas-mixed reactors achieved a higher mixing efficiency and better homogeneous mixing within a shorter mixing time than the mechanically mixed reactor.

3.2.2. Facilitated hydrolysis and VFA generation and consumption

AD is accomplished by microorganisms in different trophic groups, including hydrolysis, acidogenesis, syntrophic acetogenesis, and methanogenesis groups (Batstone et al., 2002). The initial hydrolysis of particulate complex substrates to soluble forms is a rate-limiting step during AD of complex organic matter such as animal wastewater. Proteins, carbohydrates, fiber and fat are major particulate components of swine wastewater and account for more than 50% of the TS. Therefore, changes of these particulate constituents in the solid phase can be used to describe hydrolysis efficiency. Hydrolyses of the protein, carbohydrates, fiber and fat under different digester mixing modes were shown in Table 2. As expected, mixing significantly increased the hydrolysis efficiencies with which these substances were degraded (1.6–34.7%) in comparison to the reactor without mixing (R-n) ($p < 0.05$). A uniform distribution of substrates, enzymes and microorganisms throughout the digesters during mixing played an important role in achieving the improved efficiencies. There was significant difference in hydrolysis efficiency of the protein, carbohydrates, fiber and fat between R-a and other three reactors (Dunnett's t test). The highest protein, fiber and carbohydrates removal efficiencies were obtained in R-a, for which hydrolysis efficiencies were 1.7–11.4%, 7.7–23.8% and 21.0–29.1% higher than those in R-b, R-m and R-n, respectively. The fat removal efficiency in R-a was second only to that in R-b. These results indicated that air mixing promoted the hydrolysis efficiency of particulate complex substrates. On the one hand, mixing promoted a uniform distribution of substrate and microorganisms (Singh et al., 2019). On the one hand, it was likely that injected oxygen could support the growth of and metabolism of facultative bacteria, leading to more hydrolytic enzyme production and improved the hydrolysis efficiency (Fu et al., 2016; Nguyen and Khanal, 2018). As shown by Fu et al. (2016), a daily oxygen supply obviously improved the hydrolysis rate of corn straw and some recalcitrant organic matter was also degraded.

VFAs are produced by fermentation of hydrolysis products and are used in CH_4 production. Fig. 5 illustrates the variation of VFAs concentrations (acetate, propionate and total VFA (TVFA)) in four reactors during the 24 h after feeding. Acetate concentration tended to fluctuate throughout the period, but the standard deviation of concentrations was less than 10%, which suggested that the production and consumption of acetate might have occurred simultaneously. Nevertheless, the VFA concentrations in the four reactors differed. The TVFA

Table 2
The hydrolytic efficiency in anaerobic reactors with different mixing methods.

	R-a	R-b	R-m	R-n
Proteins	17.93 ± 0.24	17.63 ± 0.36	16.65 ± 0.48	14.81 ± 0.44
Carbohydrates	15.85 ± 0.52	15.56 ± 0.52	14.48 ± 0.96	12.28 ± 0.72
Fiber	20.97 ± 0.27	19.68 ± 0.13	16.94 ± 0.15	16.68 ± 0.26
Fat	26.65 ± 0.10	23.93 ± 0.66	29.38 ± 0.20	21.81 ± 0.27

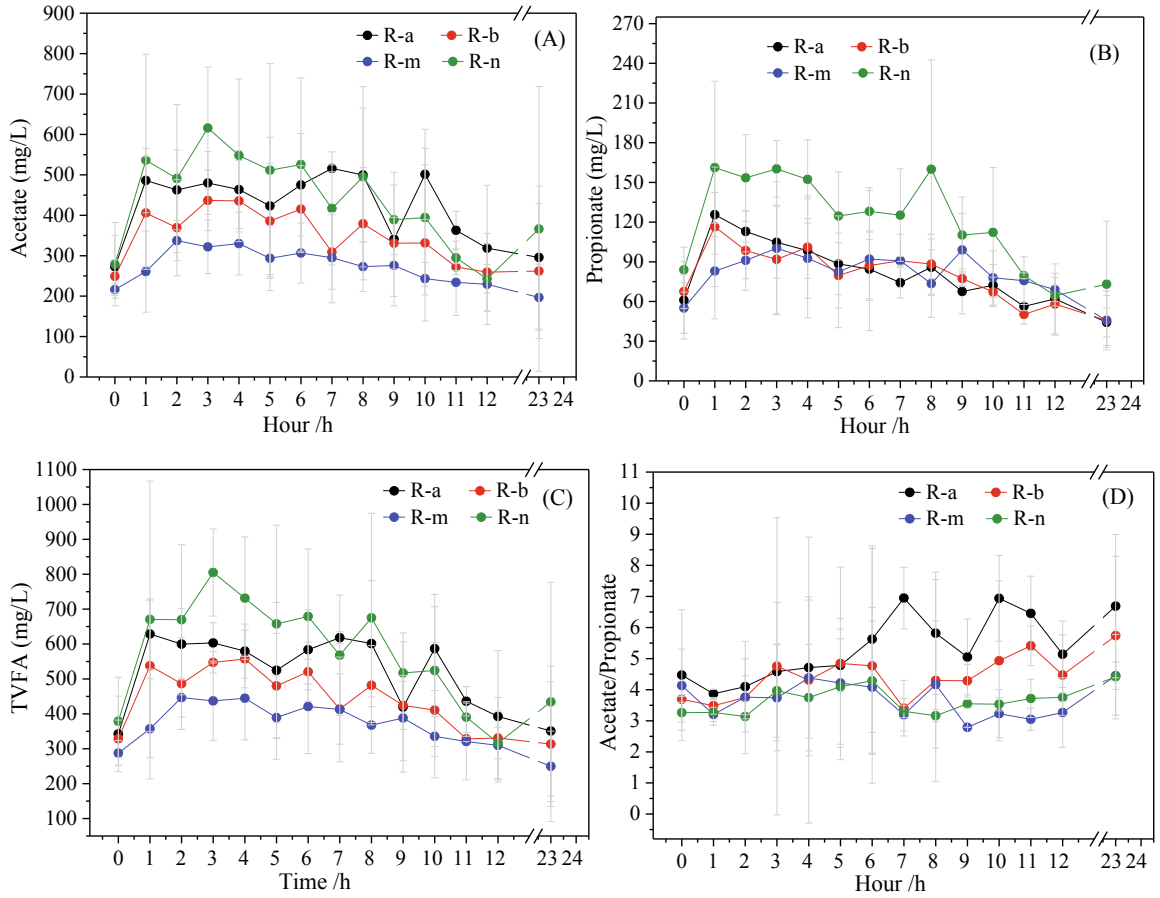


Fig. 5. Variations in concentrations of acetate (A), propionate (B) and total volatile fatty acids (TVFA) (C) and the ratio of acetate to propionate in the four anaerobic digesters over 24 h.

concentration was the highest in R-n (300–800 mg/L). This was possibly due to poor mass transfer efficiency in R-n (no mixing), resulting in an obstruction of VFA degradation by symbiotic microflora. The TVFA concentration in R-a ranged from 300 mg/L to 620 mg/L and was higher than that in R-b (300–550 mg/L) and R-m (250–450 mg/L). The greater VFAs production in R-a could be attributed to a transient microaerobic environment after air mixing, when the ORP increased to −250 mV. Alkaya and Demirer (2011) and Yin et al. (2016) showed that ORP values ranging between −100 mV and −300 mV enhanced the production of VFAs. Thus, in the present study air mixing increased the ORP to −250 mV and variations in the ORP might have promoted the formation of VFAs, thereby helping to improve the methane yield due to the production of more VFAs.

3.2.3. Optimized syntrophic relationship among the microbial communities

Bacterial and archaeal community structures at the genus level in samples from the four digesters are shown in Fig. 6. The most numerous genus *Ruminofilibacter* accounted for 19.2% of all bacteria in R-n, which was significantly higher than the abundance in three mixed reactors (6.8–11.8%) (Fig. 6 (A)). In comparison, *Treponema*_2 (instead of *Ruminofilibacter*) was the most dominant cellulose-degrading bacteria in the two gas mixed digesters (Huber and Stetter, 2015). Genus *Clostridium*_sensu_stricto_1, *Turicibacter* and vadinBC27_wastewater_sludge_group species break down saccharides or protein and produce VFAs, alcohol, CO_2 and H_2 (Ros et al., 2017; Zhang et al., 2019). Together, these three hydrolyzing bacteria in R-a and R-m were in similar abundance (23.7–23.4%) and were obviously more abundant than in R-b and R-n (16.1–19.5%). Genus norank_f_*Synergistaceae*, and norank_f_*Anaerolineaceae* were the two main syntrophic organisms, and their relative abundance was higher in three mixed reactors (7.42–9.61%) than that in the reactor without mixing (4.03%). These microorganisms may trigger the electron transfer between syntrophic bacteria and methanogens and the CO_2 reduction to CH_4 by methanogenesis.

Genera *Acinetobacter* and *Rhodococcus* were the main H_2S-oxidizing bacteria (SOB) (Pokorna and Zabranska, 2015) and their combined relative abundance in R-a was 1.2%, which was 19.6–50.6% higher than in the other three reactors (0.79–0.99) (see Supporting Information). The high abundance of SOB enhanced H_2S removal. Genera *Alcaligenes* and *Pseudomonas* were main aerobic hydrogen-oxidizing bacteria (AHOB) (Matassa et al., 2016) and their combined relative abundance in R-a was 0.11%, which was 0.6- to 1.2-fold the abundance in the other three reactors (0.05–0.07) (see Supporting Information). The high abundance of AHOB helped reduce the hydrogen partial pressure in the digester, which was conducive to the generation and conversion of VFAs to CH_4.

In air-mixed reactor (R-a), the dissolved O_2 was likely consumed by the oxidation of organic matter, H_2S and H_2 by facultative bacteria (i.e., hydrolytic and fermentative bacteria), SOB and AHOB. The fast-growing facultative heterotrophs (μ_m = 13.2/day) appeared to have an advantage over slow-growing H_2-oxidizing bacteria (μ_m = 3.6/day) and H_2S-oxidizing bacteria (μ_m = 8.6/day) (see Supporting Information). Furthermore, the Gibbs free energy changes for aerobic oxidation of VFA and H_2 ($\triangle G'_R$ = −94.0 to −99.6 kJ/e^-) was slightly higher than oxidation of H_2S ($\triangle G'_R$ = −91.8 kJ/e^-) (see Supporting Information). Consistent with this observation, facultative heterotrophs accounted for more than 30% of the total bacteria, while AHOB and SOB accounted for only 0.11% and 1.20%, respectively. Therefore, the majority of O_2 dissolved in the liquid was consumed by facultative bacteria and did not have a substantial impact on the anaerobic

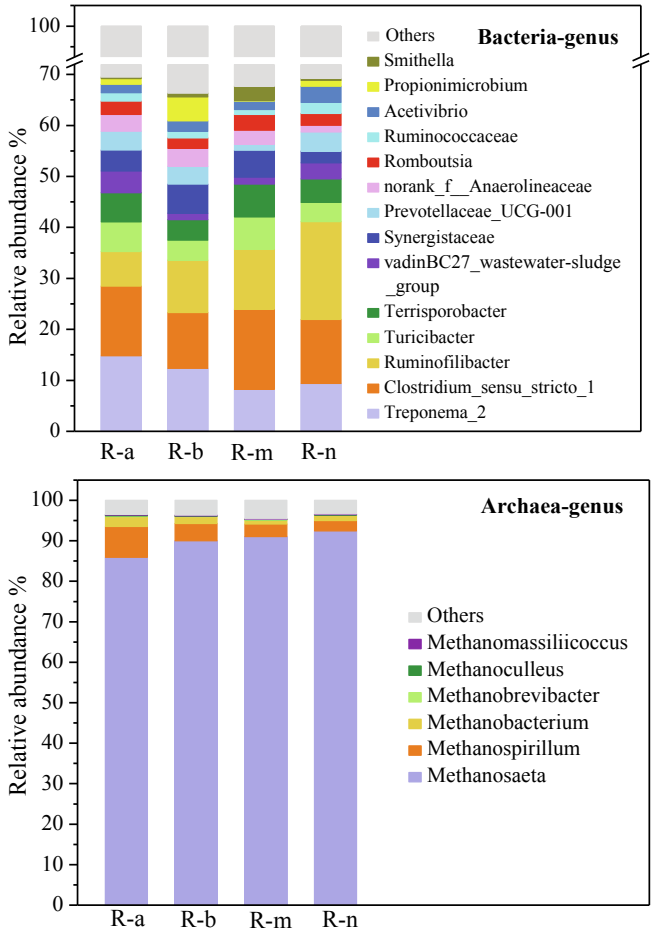


Fig. 6. Variation of bacterial and archaeal community structures at the genus level.

environment for biogas production.

Archaeal communities demonstrated a clear dominance of the genus *Methanosaeta*, which represented 86.0–92.5% of all sequences identified in the four digesters (Fig. 4 (B)). The genus *Methanospirillum* was the second most abundant genus and comprised 7.6% of the archaeal population in R-a, but its abundance reduced by half in other three reactors. In AD systems, *Methanosaeta* are known to use acetate as the substrate for methanogenesis (via the acetotrophic pathway), whereas *Methanospirillum* use the CO_2 reduction pathway with either H_2/CO_2 or formate as substrate (via the hydrogenotrophic pathway). An increase in the abundance of hydrogenotrophic methanogens such as *Methanoculleus* also was reported in several studies about microaerobic systems (Tang et al., 2004; Ramos and Fdz-Polanco, 2013). There was a good balance of acetotrophic methanogenesis and hydrogenotrophic methanogenesis in R-a and may be a reason for the high methane production compared to that in the other three reactors. In addition, the high relative abundance of *Methanospirillum* in R-a indicated that more methane could be produced via the interspecies H_2/formate transfer pathway. This pathway may be considered as a remedy for decreased methane content in biogas due to an increase of CO_2 caused by aerobic oxidation.

3.2.4. Relative contributions of the improved mass transfer and the microaerobic environment to increased methane production

The results presented thus far show that air mixing not only had no negative effect on AD, but also promoted the hydrolysis of particulate matter, allowing better solubilization for further anaerobic processes and improved methane production efficiency. Two main factors contributed to this increased efficiency. One was the improved mass transfer between the substrate and microorganisms due to mixing. The other was the transient microaerobic environment generated by the injection of air, which enabled both anaerobic and aerobic biological activities to occur within a single bioreactor. The relative contributions of the factors were analyzed based on data in Table 1. The results are presented in Table 3.

Table 3
Relative contributions of the improved mass transfer and microaerobic environment to hydrolysis and methane production.

		Mass transfer	Microaerobic environment
Hydrolysis	Proteins	91.97 ± 2.02	8.03 ± 2.02
	Carbohydrates	93.49 ± 1.17	6.51 ± 1.17
	Fiber	74.48 ± 0.47	25.52 ± 0.47
	Fat	48.88 ± 9.71	51.12 ± 9.71
Methane production		62.9 ± 1.53	37.1 ± 1.53

The volumetric methane production rates for R-a, R-b and R-n were 1.371, 1.288, 1.225 and 1.147 L/L/d, respectively (Table 1). The increase of 0.224 L/L/d in R-a (compared with the rate in R-n) was regarded as the total contribution of the improved mass transfer combined with the microaerobic environment. The increase of 0.083 L/L/d in R-a (compared with the rate in R-b) was regarded as the contribution of microaerobic environment. On this basis improved mass transfer was responsible for 62.9% of the improved methane production in R-a and the microaerobic environment was responsible for 37.1% of the improved production. Similarly, the improved mass transfer was estimated to be responsible for 48.9–93.5% of the hydrolysis efficiency of proteins, carbohydrates, fiber and fat. These results demonstrated that air mixing promoted methane production and hydrolysis more mainly by improving mass transfer than by generating a microaerobic environment.

3.3. Significance and prospect of applying air mixing in practice

The results of this study justify the conclusion that air mixing can be used as an alternative mixing method for AD and will enhance methane production. This is a new agitating approach for AD and achieves multiple functions, such as effective mixing, a creation of a microaerobic environment, the improved efficiency of hydrolyzing solid organic matter and biological desulfurization, consequently improving methane production efficiency.

Although air mixing may have a certain negative effect on the biogas composition due to dilution, the air content in the biogas in this study was approximately 10–12%, which was only slightly higher than that found by Tang et al. (2004) for the biogas that was desulfurized in the tank (2–7.5%). Less frequent mixing could further reduce the air content of the biogas. When a large amount of air is used for mixing, a three-way valve and solenoid valve can be set on the biogas outlet pipe. After air mixing, the gas discharged from the anaerobic digester can be discharged to the flare for combustion. The air content in the biogas can be completely controlled at a level as low as that in biogas desulfurization in digester. Therefore, air mixing can be fully realized in practical engineering applications of AD. Because air mixing has many advantages, it is expected that this technology has promising application prospects.

4. Conclusion

This study proposed a new mixing method for anaerobic digestion treating animal wastewater using air as gas source of agitation (named “air mixing”) and demonstrated its feasibility by comparing with other mixing strategies. Air mixing showed great advantages in hydrolysis, methane production and the removal of hydrogen sulfide in biogas. The improved mass transfer and the created microaerobic environment by

air mixing were two main factors contributing to the increased reactor performance. The proposed air mixing could be a promising option for mixing method.

CRediT authorship contribution statement

Hongnan Yang: Methodology, Writing - original draft, Writing - review & editing. **Liangwei Deng:** Conceptualization, Resources, Writing - review & editing, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2020.124047.

References

Alkaya, E., Demirer, G.N., 2011. Anaerobic acidification of sugar-beet processing wastes: Effect of operational parameters. Biomass Bioenergy 35 (1), 32–39.
Batstone, D.J., Keller, J., Angelidaki, I., Kalyuzhnyi, S., Pavlostathis, S., Rozzi, A., Sanders, W., Siegrist, H., Vavilin, V., 2002. The IWA anaerobic digestion model No 1(ADM 1). Water Sci. Technol. 45 (10), 65–73.
Bhattacharjee, P.K., Kennedy, S., Eshtiaghi, N., Parthasarathy, R., 2015. Flow regimes in the mixing of municipal sludge simulant using submerged, recirculating jets. Chem. Eng. J. 276, 137–144.
Bhattad, U., Venkiteshwaran, K., Cherukuri, K., Maki, J.S., Zitomer, D.H. 2017. Activity of methanogenic biomass after heat and freeze drying in air. Environ. Sci. Water Res. Technol., 3(1), 10.1039.C7EW00049A.
Borowski, S., Domański, J., Weatherley, L., 2014. Anaerobic co-digestion of swine and poultry manure with municipal sewage sludge. Waste Manage. 34 (2), 513–521.
Botheju, D., Bakke, R., 2011. Oxygen effects in anaerobic digestion-a review. Open Waste Manage. J. 4, 1–19.
Chen, Q., Wu, W., Qi, D., Ding, Y., Zhao, Z., 2020. Review on microaeration-based anaerobic digestion: State of the art, challenges, and prospectives. Sci. Total Environ. 710, 136388.
Cheng, Z., Wei, Y., Zhang, Q., Zhang, J., Lu, T., Pei, Y., 2018. Enhancement of surfactant biodegradation with an anaerobic membrane bioreactor by introducing microaeration. Chemosphere 208, 343–351.
Díaz, I., Lopes, A.C., Pérez, S.I., Fdz-Polanco, M., 2010. Performance evaluation of oxygen, air and nitrate for the microaerobic removal of hydrogen sulphide in biogas from sludge digestion. Bioresour. Technol. 101 (20), 7724–7730.
Díaz, I., Pérez, S.I., Ferrero, E.M., Fdz-Polanco, M., 2011. Effect of oxygen dosing point and mixing on the microaerobic removal of hydrogen sulphide in sludge digesters. Bioresour. Technol. 102 (4), 3768–3775.
Fu, S.-F., Wang, F., Shi, X.-S., Guo, R.-B., 2016. Impacts of microaeration on the anaerobic digestion of corn straw and the microbial community structure. Chem. Eng. J. 287, 523–528.
Hu, Y.-Y., Wu, J., Li, H.-Z., Poncin, S., Wang, K.-J., Zuo, J.-E., 2019. Study of an enhanced dry anaerobic digestion of swine manure: performance and microbial community property. Bioresour. Technol. 282, 353–360.
Huber, H., Stetter, K.O., 2015. Sulfolobales. Bergey's Manual of Systematics of Archaea and Bacteria, 1-1.
Kaparaju, P., Buendia, I., Ellegaard, L., Angelidakia, I., 2008. Effects of mixing on methane production during thermophilic anaerobic digestion of manure: lab-scale and pilot-scale studies. Bioresour. Technol. 99 (11), 4919–4928.
Karim, K., Hoffmann, R., Klasson, T., Al-Dahhan, M.H., 2005. Anaerobic digestion of animal waste: waste strength versus impact of mixing. Bioresour. Technol. 96 (16), 1771–1781.
Kariyama, I.D., Zhai, X., Wu, B., 2018. Influence of mixing on anaerobic digestion efficiency in stirred tank digesters: a review. Water Res., 143, S0043135418305256.
Klok, J.B.M., van den Bosch, P.L.F., Buisman, C.J.N., Stams, A.J.M., Keesman, K.J., Janssen, A.J.H., 2012. Pathways of sulfide oxidation by haloalkaliphilic bacteria in limited-oxygen gas lift bioreactors. Environ. Sci. Technol. 46 (14), 7581–7586.
Krayzelova, L., Bartacek, J., Díaz, I., Jeison, D., Volcke, E.I.P., Jenicek, P., 2015. Microaeration for hydrogen sulfide removal during anaerobic treatment: a review. Rev. Environ. Sci. Bio/Technol. 14 (4), 703–725.
Krayzelova, L., Bartacek, J., Kolesarova, N., Jenicek, P., 2014. Microaeration for hydrogen sulfide removal in UASB reactor. Bioresour. Technol. 172, 297–302.
Kuczman, O., Tavares, M.H.F., Gomes, S.D., Guedes, L.P.C., Grisotti, G., 2017. Effects of stirring on cassava effluent treatment in an anaerobic horizontal tubular pilot reactor with support medium – A Review. Renewable Sustain. Energy Rev. 77, 984–989.
Latha, K., Velraj, R., Shanmugam, P., Sivanesan, S., 2019. Mixing strategies of high solids anaerobic co-digestion using food waste with sewage sludge for enhanced biogas production. J. Cleaner Prod. 210, 388–400.
Leonzio, G., 2019. Studies of mixing systems in anaerobic digesters using CFD and the future applications of nanotechnologies. Waste Biomass Valoriz. 1–31.
Low, S. 2014. Mixing characteristics of a model anaerobic digester.
Mao, L., Zhang, J., Dai, Y., Tong, Y.-W., 2019. Effects of mixing time on methane production from anaerobic co-digestion of food waste and chicken manure: experimental studies and CFD analysis. Bioresour. Technol. 294, 122177.
Matassa, S., Verstraete, W., Pikaar, I., Boon, N., 2016. Autotrophic nitrogen assimilation and carbon capture for microbial protein production by a novel enrichment of hydrogen-oxidizing bacteria. Water Res. 101, 137–146.
Nghiem, L.D., Manassa, P., Dawson, M., Fitzgerald, S.K., 2014. Oxidation reduction potential as a parameter to regulate micro-oxygen injection into anaerobic digester for reducing hydrogen sulphide concentration in biogas. Bioresour. Technol. 173, 443–447.
Nguyen, D., Khanal, S.K. 2018. A little breath of fresh air into an anaerobic system: How microaeration facilitates anaerobic digestion process. Biotechnol. Adv., 36(7), S0734975018301459.
Nguyen, D., Wu, Z., Shrestha, S., Lee, P.-H., Raskin, L., Khanal, S.K., 2019. Intermittent micro-aeration: New strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion. Water Res. 166, 115080.
Pokorna, D., Zabranska, J., 2015. Sulfur-oxidizing bacteria in environmental technology. Biotechnol. Adv. 33 (6), 1246–1259.
Ramos, I., Fdz-Polanco, M., 2013. The potential of oxygen to improve the stability of anaerobic reactors during unbalanced conditions: Results from a pilot-scale digester treating sewage sludge. Bioresour. Technol. 140, 80–85.
Ros, M., de Souza Oliveira Filho, J., Perez Murcia, M.D., Bustamante, M.A., Moral, R., Coll, M.D., Lopez Santisima-Trinidad, A.B., Pascual, J.A., 2017. Mesophilic anaerobic digestion of pig slurry and fruit and vegetable waste: dissection of the microbial community structure. J. Cleaner Prod. 156, 757–765.
Ruan, D., Zhou, Z., Pang, H., Yao, J., Chen, G., Qiu, Z., 2019. Enhancing methane production of anaerobic sludge digestion by microaeration: enzyme activity stimulation, semi-continuous reactor validation and microbial community analysis. Bioresour. Technol. 289, 121643.
Singh, B., Szamosi, Z., Siménfalvi, Z., 2019. State of the art on mixing in an anaerobic digester: A review. Renewable Energy.
Tang, Y., Shigematsu, T., Ikbal, Morimura, S., Kida, K., 2004. The effects of micro-aeration on the phylogenetic diversity of microorganisms in a thermophilic anaerobic municipal solid-waste digester. Water Res. 38 (10), 2537–2550.
Yang, H., Deng, L., Wang, L., Zheng, D., Liu, Y., Wang, S., Huang, F., 2019. Comparison of three biomass-retaining reactors of the ASBR, the UBF and the USR treating swine wastewater for biogas production. Renewable Energy 138, 521–530.
Yin, J., Yu, X., Zhang, Y., Shen, D., Wang, M., Long, Y., Chen, T., 2016. Enhancement of acidogenic fermentation for volatile fatty acid production from food waste: effect of redox potential and inoculum. Bioresour. Technol. 216, 996–1003.
Zeng, Z., Zheng, P., Zhang, M., Ghulam, A., 2019. Performance and working mechanism of a novel anaerobic self-flotation reactor for treating wastewater with high suspended solids. Environ. Sci. Pollut. Res. 26 (25), 26193–26202.
Zhang, F., Zhang, W., Qian, D.-K., Dai, K., van Loosdrecht, M.C.M., Zeng, R.J., 2019. Synergetic alginate conversion by a microbial consortium of hydrolytic bacteria and methanogens. Water Res. 163, 114892.